DATASCI 406 FA 26, Quiz 2.3

Name	Uniqname	UMID

Each question is worth 1 point. Circle one answer for each.

Question 1 (1 point)

$X \sim$ Poisson(3) and $Y = X \mid X$ is odd. About half of Poisson(3) draws are odd. Which code produces 1000 draws of Y?

1. `x <- rpois(1000, 3); x[x %% 2 == 1]`
2. `x <- rpois(3000, 3); x[x %% 2 == 1][1:1000]`
3. `2 * rpois(1000, 3) + 1`
4. `x <- rpois(1000, 3); x[x %% 2 == 0] <- x[x %% 2 == 0] + 1; x`

Solution

Answer: 2. Rejection sampling keeps the draws that land in the set, and 3000 candidates give about 1500 odd values, of which we take 1000. Option 1 has the right distribution but only about 500 draws. Option 3 always gives an odd number, but $P(Y = 2k+1)$ is not $P(X = k)$. Option 4 moves the even values up by one instead of discarding them, so values like 1 and 3 get extra probability.

Question 2 (1 point)

You are using accept-reject to draw from the half-normal distribution, $f(x) = 2\phi(x)$ for $x \geq 0$ (the target). The candidates come from Exp(1), and $c = \sqrt{2e/\pi}$:

```
f <- function(x) 2 * dnorm(x)
g <- function(x) dexp(x)
c <- sqrt(2 * exp(1) / pi)
```

Which line of code draws 1000 candidates and keeps draws from f?

1. `y <- runif(1000); w <- runif(1000); y[w < f(y) / (c * g(y))]`
2. `y <- rexp(1000); w <- rexp(1000); y[w < f(y) / (c * g(y))]`
3. `y <- rexp(1000); w <- runif(1); y[w < f(y) / (c * g(y))]`
4. `y <- rexp(1000); w <- runif(1000); y[w < f(y) / (c * g(y))]`

Solution

Answer: 4. Each candidate needs two draws: the candidate itself from Exp(1), the distribution g in the ratio, and its own $U(0,1)$ to decide whether to keep it. Option 1 draws the candidates from $U(0,1)$, so no kept draw is above 1. Uniform candidates are right only when g is the uniform density, as in deck 3's first examples. Option 2 draws the keep-or-not number from Exp(1). The keep probability is $f(y)/c\,g(y)$ only because $P(U < r) = r$ for a uniform U, and an Exp(1) draw is below r with probability $1 - e^{-r}$. Option 3 compares every candidate with the same u, which keeps all the candidates whose ratio is above one cutoff. The kept draws then follow g cut off at that point, not f.

Question 3 (1 point)

Again using accept-reject, the target is $f(x) = 2x$ on $[0,1]$ and the candidate is $g(x) = 1$ on $[0,1]$. What is the smallest valid c, and about what fraction of candidates is accepted?

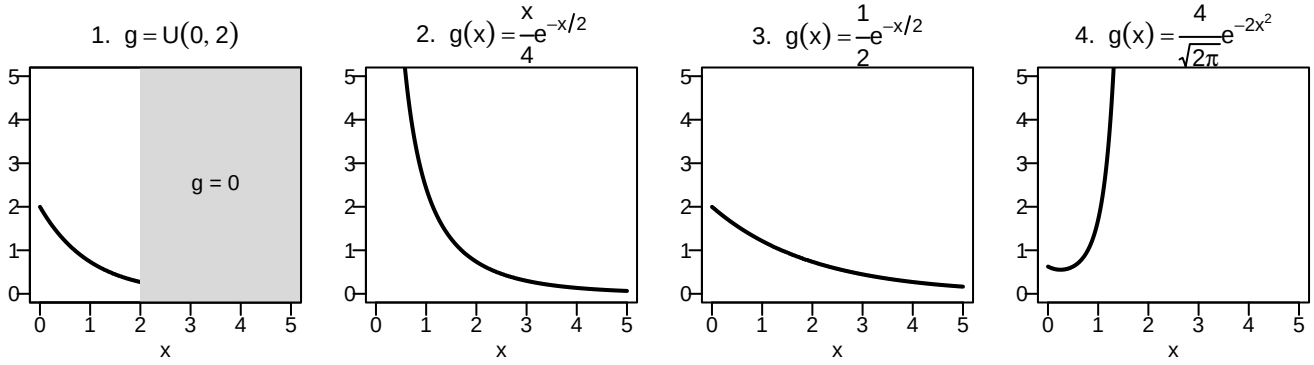
1. $c = 1$, and every candidate is accepted.
2. $c = 2$, and about half are accepted.
3. $c = 2$, and about two thirds are accepted.
4. $c = 1/2$, and about half are accepted.

Solution

Answer: 2. c must satisfy $c\,g(x) \geq f(x)$ for all x, and f is largest at $f(1) = 2$. The acceptance probability is $1/c$. Option 1 needs $f \leq g$, but $f > 1$ for $x > 1/2$. Option 3 has the right c and the wrong rate. Option 4 inverts c.

Question 4 (1 point)

Again using accept-reject, the target is Exp(1), $f(x) = e^{-x}$ for $x \geq 0$. Each panel shows the ratio $f(x)/g(x)$ for a candidate density g. Which candidate can be used to draw from f? Circle the panel number.



Solution

Answer: 3. Accept-reject needs a c with $c \geq f(x)/g(x)$ for every x. In panel 3 the ratio is $2e^{-x/2} \leq 2$, so $c = 2$ works and half the candidates are kept. In panel 1, $g = 0$ past $x = 2$ while f is still positive, so the ratio is undefined there and g never proposes those values. In panel 2 the ratio $4e^{-x/2}/x$ grows without bound as x approaches 0, because $g(0) = 0$ where f is largest. In panel 4 the ratio $\frac{\sqrt{2\pi}}{4}e^{2x^2-x}$ grows without bound as x grows, because g's tail is lighter than f's.

Rubric. 1 pt per question, no partial credit.